Rugby Taylor

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EDUCATION

Brigham Young University, Ira A. Fulton College of Engineering Apr 2027
Bachelor of Science, Electrical Engineering

- **Focus:** Power Electronics, Energy Storage, Hardware Testing
- **Certifications:** NFPA 70E (In Progress), FE Electrical and Computer Exam (In Progress)

TECHNICAL SKILLS

Power & Energy Systems: UPS systems (60-120 kVA), lithium battery modules (50V), HV interlock systems
HV Safety: LOTO, arc flash, isolation procedures, system-level troubleshooting, reliability & fault diagnosis
Analysis & Simulation: LTspice, KiCad, MATLAB, Python, GitHub Actions (CI/CD)
Embedded Systems: C/C++, ESP32, Git, real-time measurement systems, analog circuitry design
Fabrication & Prototyping: Soldering, custom hardware assembly, basic milling/machining, welding

RELEVANT EXPERIENCE

Nusano (Particle Accelerator) 190020May 2026 - Aug 2026
Electrical Systems & High Power RF Intern

- Integrated a precision sample-and-hold circuit enabling PLCs to capture high-intensity arc-flash signals previously beyond sampling limits
- Architected a multi-input interlock key system integrating 8 HV safety signals with LOTO-compliant bypass and shutdown functionality

BYU Racing Team (Formula E Car) Jan 2026 - Present
Electrical Systems Officer

- Spearheaded a full redesign of the vehicle shutdown architecture, integrating TSSI, discharge, BSPD, IMD/BMS resets, and ready-to-drive indicator logic
- Leading development of a custom 3-phase tractive inverter (100V, 15A), achieving successful switching and control on the first PCB revision
- Designed and implemented analog precharge control circuitry from first principles, establishing the team's first fully functional precharge system

Rescue Power (UPS Systems) Feb 2025 - Present
Field Engineer

- Executed installation and acceptance testing of 60–120 kVA UPS systems supporting mission-critical loads
- Refurbished 58+ high-capacity battery modules, extending system service life and improving reliability
- Diagnosed and resolved high-urgency electrical failures under time-critical conditions, restoring full UPS functionality and minimizing downtime

PERSONAL PROJECTS

Automated Signal Monitoring & Control System July 2025 - Present

- Developed an autonomous Python-based signal-analysis pipeline evaluating high-frequency data at 1Hz against multi-month historical baselines
- Designed for fully autonomous 24/7 execution and monitoring without manual intervention

Embedded Digital Oscilloscope Oct 2025 - Jan 2026

- Engineered a real-time oscilloscope platform on ESP32 with custom analog front-end and LUT-based calibration achieving <2% error with fully functioning firmware builds and CI/CD pipeline

Energy Storage System Design & Integration Aug 2023 - Jan 2024

- Designed and assembled a 10S2P, 150A peak lithium pack, reducing system cost by ~12% while maintaining thermal and electrical safety margins